

RAHUL M PALLIYALIL

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Summary

- Software engineer with 7 years of go lang experience
- Technologies: **Golang, AWS, Kubernetes, Terraform, Devops, Kafka, Lambda, GraphQL, Grpc, REST, Postgresql, Mongo**
- AWS certified solution architect
- Azure cloud adoption framework (CAF) terraform open source contributor
<https://github.com/aztfmod/terraform-azurerm-caf>

Work Experience

Curve: Backend Engineer (Contract)

(Jan 2024 – Till Date)

- Technologies - Golang, AWS, Kubernetes, REST, GraphQL, Grpc
- Curve is a digital wallet that connects all funding cards into a single debit card
- Working as a part of the curve onboarding product selector team , contributing to the development and release of new Curve app versions.
- Developed various key features in go lang including the introduction of the cooling off period concept for new Curve accounts, enabling in app downgrades for users , applying voucher based discounts for selected products, cards as an accessory etc
- Architecting and refactoring microservices to facilitate a graceful migration from product selection version 1 to version 2 and finally to version 3 for both stripe and manual billing customers.
- Migrating go lang microservices that retrieve data using complex rego expressions to mongo
- Being part of the support rota to fix production bugs based on priority.

Hackathons Participated:

- AI Based Merchant Recommendations - Participated in a company wide hackathon to recommend shopping preferences based on user transactions and won 3rd place.

Shell: Senior Software Engineer (Contract)

(Oct 2022 – Dec 2023)

- Technologies - Golang, Java, AWS, Kubernetes, Terraform, Devops, Kafka, Lambda, GraphQL, REST, Postgresql, Mongo
- Working as a part of the shell charge works fleet team for clients such as Amazon.
- Writing microservices, lambda to consume telemetry from kafka, sqs queues , enriching the data and publish enriched data to kafka
- Wrote graphql layer (including federation layer), auth workflow , subgraphs for the telematics api

- Configuring argocd, gitlab pipelines, dockerfiles and helm charts for deploying services to kubernetes and also to serverless world.
- Experience with databases such as postgres and mongo.
- Implementing service tests (godog) in golang and also extending the tests for the legacy services written in Java.

Office for National Statistics (ONS): Senior SC Cleared Software Engineer (Contract)

(Dec 2021 – Oct 2022)

- Technologies - Golang, AWS, Kubernetes, Devops, Elasticsearch, Terraform, Kafka
- Greenfield development of different microservices in golang which communicate to each other via kafka.
- Wrote a golang template for a clean architecture pattern, so that the team can use it as a baseline for the creation of new microservices.
- Part of ONS backend publishing team
- Writing facade for elasticsearch and working with the elastic analytics and search engine
- Wrote different templates for elastic search queries for efficient retrieval of the documents stored in the search engine.
- Configuring service level test framework in golang.
- Enhancing and bug fixing for the legacy applications written in Java
- Working with data driven templates written in golang.

Atos, Wokingham: Senior Software Engineer (Contract)

(Apr 2021 – Dec 2021)

- Part of Atos backend development team

Backend Development:

- Part of the portal backend team and leading portal backend development
- Writing different microservices in Golang following clean architecture patterns
- Writing RESTful apis, middlewares and focussing on test driven development
- Creating and maintaining OpenApi definitions ,documentations using stoplight
- Configuring service level test framework in go
- Building, testing and deploying to Azure kubernetes services (AKS)

Devops And Architecture:

- Being part of different architecture discussions across azure
- Written and contributed to the company adopted azure enterprise scale caf framework including the landingzone and modules.
- Written terraform to configure apim, function app, frontdoor, app gateway, cdn, database configuration etc
- Configured helm charts for packaging and argocd for deploying resources to kubernetes following an app of apps approach.

CognitoIQ, Newbury: Senior Software Engineer

(Feb 2020 – Apr 2021)

- Part of CognitioQ's platform development team
- Working for different projects ranging from field services to logistics

Backend Development:

- Architecting and writing different lambda functions in Golang following the clean architecture pattern for different projects
- Building, testing and deploying to AWS serverless world
- Writing behavior driven and integration tests in golang using docker containers
- Using Aurora serverless with MySQL and DynamoDB for persistence
- Designing databases using star schema data warehousing architecture model
- Configuring RESTful apis via AWS api gateway
- Pairing and mentoring junior team members, organizing and running company-wide backend guild

Devops Role:

- Using terraform to quickly write and deploy different AWS resources including SES configurations, AWS secret manager, code pipelines, enabling VPC flow logs, setting up of VPC peering, AWS data migration service etc
- Using AWS Cognito for authentication and authorisation
- Configuring a generic lift and shift template written in python to support golang clean layered architecture.

Certifications:

- During my time at CognitioQ, I also managed to get the AWS Certified Solutions Architect certification.

Personal Projects:

- Written a structured logging facade that uses standard logging library (eg zerolog, logrus) under the hood
- Written a golang middleware library that can be used to set request ids, perform token validation etc before reaching the target lambda
- Written a distributed tracing facade which uses opencensus and with different backend exporters

Kaluza (OVO Energy), London: Software Engineer

(Nov 2019 –Feb 2020)

- Part of Kaluza's device integration team, integrating various devices to the grid
- Written Golang micro services to ingest and enrich telemetries and collect data.
- Test driven development using testify Mock, GoMock etc.
- Knowledge and experience working on Google cloud infrastructure
- Using datastore, elastic and storage buckets for persistence.
- Configured Logstash to pick up the events from Kafka, filter and push to ES
- Initiating communication between services using Kafka
- Configured modules, ACL for topics etc using terraform
- Written tests using frameworks such as GoDog, Gock, etc
- configuring monitoring and alerting tool like Prometheus and configuring Graphana dashboards.

On the Dot, London: Backend Engineer

(Apr 2019 – Nov 2019)

- Supporting and maintaining On the Dot's store to door delivery platform
- Written Golang micro services to initiate order booking, determine capacity in region for a specific order etc.
- Emphasizing on clean architecture like Layered, Hexagonal etc and test driven development
- Knowledge and experience working on AWS infrastructure
- Using dynamoDB and s3 buckets for persistence.
- Initiating communication between services using AWS SNS and SQS.
- Experience with git version control.
- Experience writing service level tests and acceptance tests
- Experience with logging dashboard like Humio and tracing with Opensensus.
- Writing and improving on Restful apis, using Swagger for api definitions.

Schlumberger, Abingdon: Software Engineer (Full Stack)

(2016 – Apr 2019)

Backend stack:

- Written Golang micro services for gathering data, running simulations and run calculations.
- Configuring and Deploying services to GCP
- A mix of both Choreography and Orchestration architecture is used.
- Focussing on test driven development and go clean architecture
- Taking part in team software development architecture meetings.
- Developing and Maintaining Restful apis (inc. use of swagger)
- Experience writing microservices for Google cloud platform
- Experience with Google PubSub messaging service
- Experience with Google protocol buffers, storage FUSE.
- Using Google datastore and storage buckets for persistence
- Experience with docker and kubernetes
- Experience with git version control.
- Experience with other technologies like node, websockets
- Experience writing gauge api tests

Frontend stack:

- Experience in writing angular 6 typescript front end applications
- Experience with jasmine testing framework with karma test runner
- Focussing on test driven development

Devops:

- Creating groovy tasks for gradle build
- Running static code and dynamic api scans for different microservices
- Knowledge and experience in configuring builds on gocd and vsts pipelines

Security:

- Knowledge and experience in implementing oauth2 protocol

Hackathons Participated:

- Let's Monkey About - Injecting chaos into a running kubernetes cluster and monitoring the

performance , load balancing, finding possible solutions etc.

- Go Delve Debugging - Debugging multiple docker go containers, which communicate via pubsub and REST.
- License Dashboard - Dashboard service created in Angular 2 that displays licensing information.

Schlumberger, Abingdon: Build and Release (Baseline) Engineer

(Dec 2013 – 2016)

Roles Include:

- Experience with cross platform continuous integration and delivery tools
- Configuring builds via continuous integration tool TeamCity, Microsoft TFS
- Writing Automated scripts mainly in python
- Deploying to Google cloud platform /Google app engine via Maven and gcloud.
- Running simulations on Microsoft Azure cloud.
- Maintaining and Supporting Perforce version control system
- Working on both windows and linux environment
- Experience with build tools like Make, CMake, Ant, Maven and SBT(Simple Build Tool)
- Maintaining and supporting Teamcity Build scripts in python, MSBuild
- Configuring customized reports via Report Builder
- Working with SQL Server Management Studio
- Technical analysis with large clients which involves running meetings on a regular basis
- Sending out Technical review status mail to teams and clients
- Follow up on actions based on Technical review meeting
- Running tests suits and making sure these are ready for release
- Configuring and Running tests on Microsoft Azure cloud, CentOS
- Experience working in automated deployment environment
- Updating status via Microsoft TFS
- Experience with Microsoft Visual Studio
- Experience with developing a Mean app with Angular2.0 Typescript
- Experience with using MongoDB as a part of the MEAN stack.
- Testing and Adding different tests to Perforce
- Personal Project Validation code in C++

Achievements:

- Deployed simple Spring mvc web api to Google app engine
- Written fully automated build script in python for library publish
- Configured tests to be submitted and run from cloud
- Presentations, Webinars and articles on various topics such as Continuous integration, code reviews via crucible
- Experience working with automating deployments
- Written python scripts to enhance build performance

Markit, London: Associate Build Engineer

(May 2013– Sep 2013)

Roles Include:

- Maintaining Buildbot and configuring builders using python programming

- Writing build scripts in python
- Personally responsible for ensuring all the builds are green
- Experience with GNU make.
- Maintaining Perforce Version Management System, carried out an upgrade to latest 2013.1 perforce service
- Creating various perforce triggers to do pre-commit check on specific files
- Converting perl scripts to python
- Basics of Bash Scripting

MyDrive Solutions, Guildford: Graduate Software Engineer

(Nov 2012 – Mar 2013)

Roles Include:

- Working within Ruby on Rails and Sinatra web framework
- Writing and Fixing Rails RSpec tests cases.
- Test Driven Development
- Using RStudio to generate plots and graphs based on data through csv.
- Data analysis on research projects.
- Creating a visualisation tool for checking efficiency.
- Committing changes to Github.

Educational Qualification

Cardiff Metropolitan University, UK: MBA, International business – 2:1

Mahatma Gandhi University, India: B Tech, Computer Science & Engineering – 2:1

Certifications

AWS Certified solutions Architect - Associate